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COUPLED STALENESS CLOCKS IN FACTOR MODELS: MULTIVARIATE INVERSE SUBORDINATION AND ITS EMPIRICAL BOUNDARY

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We study a multi-asset factor model in which each latent Markov factor is observed through its own staleness clock, given by the undershoot of a component of a multivariate subordinator with common-shock dependence. This addresses an open problem in the additive-subordination literature: time changes generated by the inverse rather than the direct subordinator. We show that the vector of normalized undershoots is exactly self-similar, that its marginals are blind to the common shock, and that common shocks create an atom of simultaneous staleness for which we derive a compensation-formula representation in terms of the bivariate potential measure. For stable index one half we obtain that measure in closed form and establish its logarithmic diagonal ridge. We derive an exact local evolution equation driven by size-biased mixing measures, show that it collapses to a classical confluent hypergeometric identity for a single clock, and quantify the failure of the marginal operators to tensorize. Under staleness alone the correlation term structure is exactly flat, and across a large parameter sweep its shape is not identified. Finally we test the load-bearing tail assumption on cryptocurrency and Vietnamese equity trade data. The estimated tail index turns out to be a property of the event definition and the diurnal cycle rather than of the market, and no series is both a plausible power law and infinite-mean.

Keywords: Price staleness; inverse subordinator; additive subordination; factor models; inter-trade durations; Marshall–Olkin dependence.

Mathematics Subject Classification 2020: 60G51, 60G22, 91G70

1. Introduction

Factor models organize the cross-section of asset returns around a small number of latent drivers. Two research programs have recently pushed those drivers away from the Markov property. The first is motivated by the statistical behavior of volatility and replaces Brownian drivers by processes with memory (Gatheral *et al.* 2018). The second is motivated by market microstructure: trades arrive at irregular, non-exponentially spaced times, and between trades a recorded price does not move at all, so observed prices are *stale* (Lo & MacKinlay 1990, Bandi *et al.* 2017). The mechanism that produces staleness is a random, singular time change.

This paper belongs to the second program and has two halves. The first develops the multivariate theory the literature has explicitly asked for. The second tests the assumption that theory rests on, and finds it unsupported. We report both, because the negative result is the more consequential of the two for anyone planning to build in this area.

1.1. *Direct and inverse subordination*

Time changes entered finance through Clark (1973), who ran a Brownian motion on a random business clock, and through the Lévy constructions of Geman *et al.* (2001). When the clock is a subordinator S the time-changed process $Y(S(t))$ remains Markov. Direct subordination speeds time up: it produces heavy tails and volatility clustering, but never freezes the price.

Staleness requires the opposite operation. Let

$$L(t) := \inf\{u > 0 : S(u) > t\}, \qquad H(t) := S\big(L(t)-\big), \tag{1.1}$$

so L is the first-passage process of S and $H(t)$, the *undershoot*, is the largest point of the range of S at or below t. For a stable subordinator that range is a Lebesgue-null regenerative set, so H is a devil's staircase: constant off a null set. Reading the range of S as the set of trade times, $H(t)$ is the timestamp of the last trade before t, and $Y(H(t))$ is the recorded price – frozen between trades, updated at trades. The staleness is $t - H(t)$. Processes time-changed this way are not Markov and their laws solve nonlocal equations (Meerschaert & Scheffler 2004, Toaldo 2015, Meerschaert & Sikorskii 2011); this is the continuous-time random walk limit that Scalas *et al.* (2000) and Mainardi *et al.* (2000) brought into finance.

1.2. *The gap this paper addresses*

Two recent papers bracket the problem. Ascione *et al.* (2024) take one asset and one staleness clock and derive the coupled nonlocal equations for a Markov process time-changed by the undershoot of a general subordinator. D'Onofrio *et al.* (2025) take k factors and k clocks, but their clocks are *direct* additive subordinators, so the resulting multiparameter process stays Markov and no staleness arises; their concluding section names time changing by the *inverse* of an additive subordinator as the open next step.

We take that step. Section 2 defines a k-factor model in which factor j is observed through the undershoot H_j of a subordinator $S_j = X_j + a_j Z$, with X_j idiosyncratic and Z a common shock. The common shock is what makes the problem genuinely multivariate: it makes the clocks dependent, and it makes two assets go stale at the same time.

1.3. *Contributions*

We state at the point of first use which results are classical. Four objects below have classical antecedents and say so in their own statements rather than in a concluding caveat.

1. **Exact self-similarity of the clock vector** (Proposition 3.1). The law of $t^{-1}(H_1(t), \ldots, H_k(t))$ on $[0,1]^k$ does not depend on t. Its marginals are the classical Dynkin–Lamperti generalized arcsine law (Lamperti 1958, Dynkin 1961, Bertoin 1999); what is new is that common shocks leave those marginals *exactly* unchanged (Proposition 3.2) while changing the joint law completely. Marginal duration data therefore cannot identify the common shock at all.
2. **An atom of simultaneous staleness, made computable** (Proposition 3.4, Theorem 3.1). With positive loadings the event $\{L_1(t) = L_2(t)\}$ has positive, t-free probability: both assets are frozen over a common interval. That common shocks generate Marshall–Olkin structure through first-passage times is known (Marshall & Olkin 1967, Sun *et al.* 2017). What we add is a compensation-formula representation of the atom in terms of the bivariate potential measure, which turns it into a quantity one can evaluate.
3. **Closed-form bivariate potential density at** $\alpha = 1/2$ (Theorem 3.2), homogeneous of degree $\alpha - 2$, with a logarithmic ridge on the diagonal (Corollary 3.1). This is the input the previous item requires.
4. **An exact local evolution equation** (Theorem 4.1): $\partial_t q = \sum_j \alpha_j G_j q^{(j)}$, where $q^{(j)}$ is built from the mixing measure size-biased in its jth coordinate. The time change is singular, so a local equation is not available by differentiating through it (Remark 4.1); it comes from the exact scaling instead. For $k = 1$ the equation collapses to the derivative identity for the confluent hypergeometric function (Proposition 4.2) – textbook, and we show the collapse explicitly so that nobody mistakes the univariate case for a new result. We also quantify the failure of the marginal operators to tensorize (Proposition 4.3).
5. **The correlation term structure carries no information.** Under staleness alone – Brownian factors – the correlation between two stale assets is *exactly* constant in the return horizon (Proposition 5.1). Adding mean reversion produces a term structure, but across 139,968 parameter combinations all four monotonicity classes occur with frequencies between 21.1% and 31.4% (Section 5). A slope-sign rule we had conjectured classifies only 52.6% of cells, which is chance. We report this because it removes a headline claim, and because a reader who builds on the model needs to know it.
6. **An empirical boundary** (Section 6). The construction requires the inter-trade duration tail index to satisfy $\alpha < 1$. On 21 cryptocurrency pairs spanning 5 orders of magnitude in turnover and 30 Vietnamese equities, the estimated index turns out to be a property of the *event definition* and the *diurnal cycle*, not of the market: the median Hill index rises from 0.24 when individual fills are counted as events to 1.19 when orders are, and to 1.61 after merging within one second.

Removing the diurnal cycle raises it further. Once the tail threshold is fitted properly, *no* series in either market is both a plausible power law and infinite-mean.

7. **The one prediction that survives, tested** (Section 6.7). Because marginal durations are blind to the common shock and the correlation term structure is flat, the only observable signature left is co-trading dependence. It is there, and it survives stratifying out the diurnal cycle: the Mantel–Haenszel odds ratio is 1.03 on Binance and 1.13 on HOSE, above one for 80% and 76% of asset pairs. But it is small – smaller than the 1.37–1.87 the same statistic takes inside the model with *zero* common-shock loading, because an α-stable clock is bursty at every scale. The framework is over-strong in dependence as well as in tails.

1.4. *What the negative result implies*

The two halves are not in contradiction, and Section 7 separates them. Results that depend on *dependence* – the simultaneous-staleness atom, its potential-measure representation, the failure of tensorization – are indifferent to the value of α and survive. Results that depend on *self-similar scaling* – the exact Beta mixing law and the local evolution equation – do not.

The data point to a specific repair. Across Vietnamese equities the tail index sits in a band around 1.75 while the median inter-trade gap ranges from 1 to 44 seconds; illiquidity moves the *scale* of durations and leaves the tail alone. A tempered stable clock, with a common index α and an asset-specific tempering rate θ_j, reproduces exactly that: it behaves like an α-stable clock below the horizon $1/\theta_j$ and has a finite mean above it. Staleness then enters as a crossover at an estimable time scale rather than as a self-similar fraction of every horizon.

Sections 2–5 develop the theory, Section 6 the empirical test, Section 7 the consequences. Proofs are in Appendix A, simulation and estimation methods in Appendix B, and reproduction instructions in Appendix C. Every number reported below is produced by the accompanying code and regenerated automatically into this manuscript.

2. The Model

2.1. *Clocks with common shocks*

Fix a filtered probability space carrying independent standard stable subordinators $X_1, \ldots, X_k$ and Z, with

$$\mathbb{E}\big[e^{-\lambda X_j(u)}\big] = e^{-u\lambda^{\alpha_j}}, \qquad \mathbb{E}\big[e^{-\lambda Z(u)}\big] = e^{-u\lambda^{\alpha_c}}, \qquad \alpha_j, \alpha_c \in (0,1). \tag{2.1}$$

The Lévy measure of a standard α-stable subordinator is $\Pi_\alpha(dz) = \frac{\alpha}{\Gamma(1-\alpha)} z^{-1-\alpha} dz$, with tail

$$\bar{\Pi}_\alpha(z) := \Pi_\alpha\big((z,\infty)\big) = \frac{z^{-\alpha}}{\Gamma(1-\alpha)}, \qquad z > 0. \tag{2.2}$$

Definition 2.1. The *clock vector* is the k-dimensional subordinator $S = (S_1, \ldots, S_k)$,

$$S_j(u) := X_j(u) + a_j Z(u), \qquad a_j \geq 0, \quad j = 1, \ldots, k, \tag{2.3}$$

with inverse and undershoot defined coordinatewise as in (1.1).

Each S_j has Laplace exponent $\psi_j(\lambda) = \lambda^{\alpha_j} + a_j^{\alpha_c} \lambda^{\alpha_c}$, so S_j is itself stable exactly when $\alpha_j = \alpha_c$ or $a_j = 0$. The components are dependent through Z alone: their jumps coincide precisely at the jump times of Z, where $\Delta S_j = a_j \Delta Z$. This is the multivariate-subordinator construction of Barndorff-Nielsen *et al.* (2001) specialized to the factor structure of Semeraro (2008) and Luciano & Semeraro (2010), and it is the inverse-clock counterpart of the additive subordination of D'Onofrio *et al.* (2025).

2.2. *Observed prices*

Let $Y_1, \ldots, Y_k$ be independent conservative Feller processes on $\mathbb{R}$ with generators G_j and transition densities p_j, independent of the clocks. The leading example is the Ornstein–Uhlenbeck factor

$$dY_j(u) = -\kappa_j \, Y_j(u) \, du + \sigma_j \, dW_j(u), \qquad \kappa_j \geq 0, \ \sigma_j > 0, \tag{2.4}$$

with $\kappa_j = 0$ giving Brownian motion.

Definition 2.2. The *observed factors* are $\tilde{Y}_j(t) := Y_j(H_j(t))$, and the observed log-price of asset i is $P_i(t) = \sum_j b_{ij} \, \tilde{Y}_j(t)$.

Because H_j increases only on the range of S_j, each $\tilde{Y}_j$ is constant off a Lebesgue-null random set. Figure 1 illustrates one clock; Table 2.2 places the construction in the literature.

3. The Joint Law of the Staleness Clocks

Assumption 3.1. $\alpha_1 = \cdots = \alpha_k = \alpha_c =: \alpha \in (0, 1)$ *and* $a_j > 0$ *for all* j.

Table 1. Time-changed factor constructions. "Markov" refers to the time-changed process in calendar time.

Construction	Clock	Factors	Markov	Staleness
Clark (1973), Geman *et al.* (2001)	S	1	yes	no
Barndorff-Nielsen *et al.* (2001), Semeraro (2008)	S (multivariate)	k	yes	no
D'Onofrio *et al.* (2025)	additive, multiparameter	k	yes	no
Ascione *et al.* (2024)	H (undershoot)	1	no	yes
This paper	$H_1, \ldots, H_k$, common shocks	k	no	yes

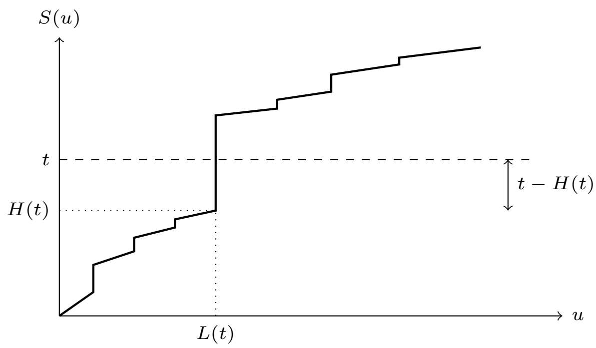

Fig. 1. One staleness clock. The subordinator S crosses the calendar level t at operational time $L(t)$; the undershoot $H(t) = S(L(t)-)$ is the last trade time at or before t, and $t - H(t)$ is the staleness. The recorded price is the efficient price at $H(t)$, so it is frozen for the whole straddling jump.

Table 2. Self-similarity of the joint clock law over a 64-fold range of horizons ($t \in \{0.25, 1, 4, 16\}$), 40,000 paths per cell. The last column is the smallest two-sample Kolmogorov–Smirnov p-value over eight random projections of the joint law at the extreme horizons.

α	$\operatorname{corr}(H_1/t, H_2/t)$ min	$\operatorname{corr}(H_1/t, H_2/t)$ max	$\Pr[L_1(t) = L_2(t)]$ min	$\Pr[L_1(t) = L_2(t)]$ max	two-sample p (joint)
0.3	0.3370	0.3498	0.3053	0.3170	0.159
0.4	0.3494	0.3541	0.2929	0.2973	0.008
0.5	0.3626	0.3736	0.2738	0.2764	0.276

3.1. *Exact self-similarity*

Proposition 3.1. *Under Assumption 3.1, for every $c > 0$,*

$$\big(H_j(ct)\big)_{j\le k} \overset{d}{=} c\,\big(H_j(t)\big)_{j\le k}, \qquad \big(L_j(ct)\big)_{j\le k} \overset{d}{=} c^{\alpha}\,\big(L_j(t)\big)_{j\le k}, \tag{3.1}$$

jointly. Hence the law μ_α of $t^{-1}(H_1(t), \ldots, H_k(t))$ on $[0,1]^k$ does not depend on t.

Table 3.1 reports the test. Over a 64-fold range of horizons, and for three values of α, both the pairwise correlation of the normalized undershoots and the simultaneity probability of Section 3.4 are constant to within Monte Carlo error.

3.2. *Marginals are blind to the common shock*

Proposition 3.2 (classical, with a common-shock corollary). *Let S be a standard α-stable subordinator, $\alpha \in (0,1)$. Then for every $t > 0$,*

$$\frac{H(t)}{t} \sim \mathrm{Beta}(\alpha,\, 1-\alpha), \qquad \textit{density}\ \ \frac{\sin(\pi\alpha)}{\pi}\, s^{\alpha-1}(1-s)^{-\alpha} \ \ \textit{on}\ (0,1). \tag{3.2}$$

Under Assumption 3.1 the same holds for each H_j of Definition 2.1, for every $t > 0$ and every $a_j \geq 0$.

The first statement is the Dynkin–Lamperti generalized arcsine law (Lamperti 1958, Dynkin 1961); see Bertoin (1999, Sec. 3) for the subordinator form. It is not a contribution of this paper. The second is immediate: under Assumption 3.1 the sum $X_j + a_j Z$ of independent α-stable subordinators is again α-stable, with exponent $(1 + a_j^{\alpha})\lambda^{\alpha}$, and (3.2) is scale invariant.

The economic content is sharp. The common shock is *invisible* in any single asset's staleness distribution: it can only be detected jointly. Any calibration strategy based on marginal duration data is therefore blind to a_j, and the simultaneity atom of Section 3.4 is the natural thing to calibrate against instead.

Remark 3.1. If $\alpha_j \neq \alpha_c$ then S_j is not stable, $H_j(t)/t$ depends on t, and (3.2) holds only in the limit: ψ_j is regularly varying at 0 with index $\beta_j = \min(\alpha_j, \alpha_c)$, so Dynkin–Lamperti gives $H_j(t)/t \Rightarrow \mathrm{Beta}(\beta_j, 1-\beta_j)$ as $t \to \infty$. Every exact statement below fails in that case; only the asymptotic ones survive.

Remark 3.2 (the marginal law as a simulator control). Because (3.2) is exact, comparing simulated $H_j(t)/t$ with $\mathrm{Beta}(\alpha, 1-\alpha)$ validates the simulator rather than the model. Across 24 such tests the smallest Kolmogorov–Smirnov p-value is 0.007 and the largest absolute error in the mean is 0.0032 against an exact value of α. The simulator is described in Appendix B; it enumerates jumps exactly and never discretizes time.

3.3. *The Kummer transform*

Proposition 3.3. *Under the hypotheses of Proposition 3.2, for every $c \in \mathbb{R}$,*

$$\mathbb{E}\big[e^{-cH(t)}\big] = {}_1F_1(\alpha; 1; -ct), \tag{3.3}$$

and for the Ornstein–Uhlenbeck factor (2.4) *started at y,*

$$\mathbb{E}\big[\tilde{Y}_j(t) \,\big|\, Y_j(0) = y\big] = y\, {}_1F_1(\alpha; 1; -\kappa_j t). \tag{3.4}$$

Equation (3.3) is the moment generating function of a $\mathrm{Beta}(a, b)$ variable, ${}_1F_1(a; a+b; \cdot)$, at $a = \alpha$, $b = 1 - \alpha$ (Olver *et al.* 2010, Sec. 13); it is stated because (3.4) is where ${}_1F_1$ enters the model, and because Section 4 shows that the whole univariate evolution equation is a restatement of a derivative identity for this function. Numerically, ${}_1F_1$ and direct quadrature of the Beta integral agree to a relative 6.2×10^{-14}, and simulation matches to within 2.46 standard errors.

3.4. *Simultaneous staleness*

Proposition 3.4. *Under Assumption 3.1 with $k \geq 2$, for $j \neq h$,*

$$\pi_{jh} := \mathbb{P}\big[L_j(t) = L_h(t)\big] \in (0,1) \qquad \textit{and does not depend on } t > 0. \tag{3.5}$$

Table 3. The simultaneity atom π_{12} at $\alpha = 1/2$ against the compensation-formula prediction (3.6). 60,000 exact paths per row; loadings $a_1 = a_2 = a$.

a	simulation	s.e.	Eq. (3.6)	difference	z
0	0.00000	0.00000	0.00000	+0.00000	—
0.25	0.16625	0.00152	0.16370	+0.00255	+1.68
0.5	0.21340	0.00167	0.21363	-0.00023	-0.14
1	0.27325	0.00182	0.27274	+0.00051	+0.28
2	0.33950	0.00193	0.33977	-0.00027	-0.14
4	0.41400	0.00201	0.41234	+0.00166	+0.83

Note: Crossings are compared by event index, so $\{L_1 = L_2\}$ is decided exactly and the atom cannot be a discretization artifact. The residual quadrature error in the theoretical column is about 0.00021.

If $a_j a_h = 0$, then $\mathbb{P}[L_j(t) = L_h(t)] = 0$.

The event says both clocks cross the calendar level t at the same operational time, which requires a jump common to both and hence a jump of Z; both prices are then frozen over the same calendar interval. That multivariate subordinators generate Marshall–Olkin structure through first-passage times is established in Sun *et al.* (2017), building on Marshall & Olkin (1967). What we add is the following, which makes π_{jh} computable.

Theorem 3.1. *Let $U_{jh}(ds_j, ds_h) := \int_0^\infty \mathbb{P}[(S_j(u), S_h(u)) \in (ds_j, ds_h)]\, du$ be the bivariate potential measure of (S_j, S_h). Then*

$$\pi_{jh} = \int_{[0,t]^2} \bar{\Pi}_{\alpha_c}\left(\frac{t - s_j}{a_j} \vee \frac{t - s_h}{a_h}\right) U_{jh}(ds_j, ds_h). \tag{3.6}$$

The proof (Appendix A) applies the compensation formula: conditionally on the pre-jump position (s_j, s_h), a common jump ΔZ straddles t in both coordinates exactly when $a_j \Delta Z > t - s_j$ and $a_h \Delta Z > t - s_h$.

Table 3.4 tests (3.6) against exact simulation, at $\alpha = 1/2$ where the potential density of Theorem 3.2 below is available in closed form. Agreement is within 1.68 standard errors at every loading, and the atom vanishes identically when the loading does. At $a = 1$ simulation gives 0.27325 ± 0.00182 against a predicted 0.27274. Over a 64-fold range of horizons the estimate varies by 0.0054 around a pooled 0.2747 ($\chi^2 = 3.71$ on 3 degrees of freedom), as Proposition 3.4 requires.

3.5. *The bivariate potential density*

Proposition 3.5. *Under Assumption 3.1 the measure U_{jh} is absolutely continuous off the diagonal, with density satisfying*

$$u(cs_j, cs_h) = c^{\alpha-2}\, u(s_j, s_h), \qquad c > 0. \tag{3.7}$$

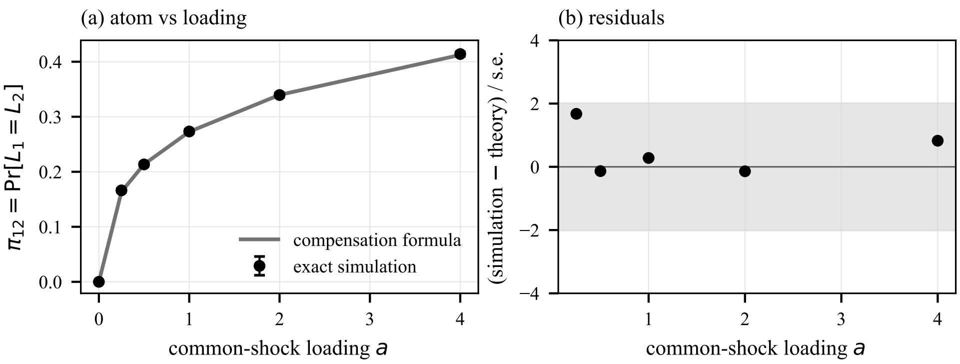

Fig. 2. Left: the simultaneous-staleness atom against the common-shock loading, exact simulation (markers, two standard errors) versus the compensation formula (3.6) evaluated with the closed-form potential density (line). Right: residuals in units of the simulation standard error; the shaded band is ± 2.

Theorem 3.2. *Let $\alpha = 1/2$, $a_j = a_h = a > 0$, with X_j, X_h, Z standard $\frac{1}{2}$-stable. Writing $x_j = s_j - az$ and $x_h = s_h - az$, for $s_j, s_h > 0$*

$$u(s_j, s_h) = \pi^{-3/2} \int_0^{(s_j \wedge s_h)/a} \frac{\sqrt{z\, x_j\, x_h}}{\big(x_j x_h + z\,(x_j + x_h)\big)^2}\, dz. \tag{3.8}$$

The derivation (Appendix A) uses the explicit $\frac{1}{2}$-stable density and evaluates the operational-time integral in closed form. Three checks are worth recording. Expression (3.8) is homogeneous of degree $-3/2 = \alpha - 2$ as Proposition 3.5 requires, verified numerically to a relative 6.7×10^{-16} – machine precision, as befits an identity. Substituting it into (3.6) yields a t-free π_{jh}, as Proposition 3.4 requires. And it reproduces the simulated atom (Table 3.4).

Corollary 3.1. *As $s_h \to s_j$ the density* (3.8) *diverges logarithmically: $u(s, s+\delta) = -\,\pi^{-3/2} c(s) \log \delta + O(1)$ as $\delta \downarrow 0$, with $c(s) > 0$.*

Over 5 decades, $u(1, 1+\delta)$ rises from 0.1306 to 0.6550 in increments of 0.1049 ± 0.0022 per decade: a constant increment per decade is the signature of a logarithmic divergence (Figure 3, left).

3.6. *Resolving the diagonal*

Comparing (3.8) with the exact occupation measure cell by cell over 144 cells, agreement away from the diagonal is good: the ratio of closed form to simulation has median 1.005 with a 5–95% range of 0.882–1.148 and a mean absolute deviation of 5.7%.

On the diagonal, plain tensor quadrature of the cell fails badly, giving ratios between 1.042 and 1.217 (median 1.134). The cause is Corollary 3.1: the logarithmic

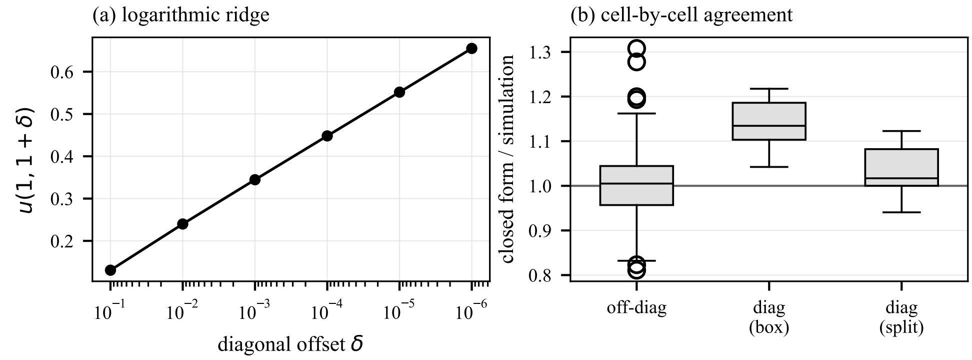

Fig. 3. Left: the logarithmic ridge of the bivariate potential density at $\alpha = 1/2$; a constant increment per decade in δ indicates a logarithmic divergence. Right: ratio of the closed form (3.8) to the exact occupation measure, cell by cell. Off the diagonal the two agree. On the diagonal, plain tensor quadrature fails; splitting the cell along the ridge repairs it.

ridge runs through the cell interior, and symmetric Gauss–Legendre nodes cannot resolve it. Cutting each diagonal cell along $s_j = s_h$ and integrating the two triangles separately, with a substitution that annihilates the logarithm at the ridge, moves the ratios to 0.941–1.123 (median 1.017); see Figure 3, right.

This settles a question left open in earlier work on this model, where a diagonal discrepancy of the same kind had been observed and it was not known whether it indicated a quadrature failure or an error in the derivation. It is a quadrature failure.

4. Evolution Equations

Write $q(t,\cdot)$ for the density of $(\tilde{Y}_1(t), \ldots, \tilde{Y}_k(t))$.

Proposition 4.1. *Under Assumption 3.1, for every $t > 0$,*

$$q(t, y_1, \ldots, y_k) = \int_{[0,1]^k} \prod_{j=1}^{k} p_j(ts_j, y_j)\ \mu_\alpha(ds), \tag{4.1}$$

with μ_α the t-free law of Proposition 3.1.

The factors are independent only *conditionally* on the clocks; the unconditional law is not a product, because μ_α is not a product measure when the loadings are positive.

Remark 4.1. It is tempting to differentiate through the time change, writing $\partial_t p_j(H_j(t), y_j) = \dot{H}_j(t) G_j p_j$. This is wrong: the range of S_j is Lebesgue-null, so $\dot{H}_j = 0$ almost everywhere and the right-hand side vanishes. It is exactly this singularity that forces the nonlocal formulation of Ascione *et al.* (2024) and Toaldo (2015). Theorem 4.1 obtains a local equation nonetheless – not by differentiating the time change, but by exploiting the exact scaling in (4.1).

4.1. *An exact local equation*

For a probability measure μ on $[0,1]^k$ with $\int s_j \mu(ds) = m_j$, write $\mu^{(j)}(ds) := s_j\,\mu(ds)/m_j$ for its *size-biased* version in coordinate j, and let $q^{(j)}$ be obtained by substituting $\mu_\alpha^{(j)}$ for μ_α in (4.1).

Theorem 4.1. *Under Assumption 3.1, $m_j = \alpha$ for every j and*

$$\partial_t\, q(t,\cdot) \;=\; \sum_{j=1}^{k} \alpha\, G_j\, q^{(j)}(t,\cdot), \qquad t>0, \tag{4.2}$$

with G_j acting on the jth coordinate.

The size-biasing constant is the coordinate mean of μ_α, which Proposition 3.2 fixes at exactly α; simulation gives 0.49776, within 1.55 standard errors. Testing (4.2) against the test function $\varphi(y) = y_1 y_2$ turns both sides into the joint Laplace transform of the clock vector (Appendix B); the ratio of right- to left-hand side lies in $[0.9763, 1.0093]$ across five parameter sets, with a largest discrepancy of 1.21 standard errors (Table 4.2).

Proposition 4.2. *For $k = 1$ with the Ornstein–Uhlenbeck factor* (2.4)*, equation* (4.2) *evaluated on the first moment is exactly the derivative identity*

$$\frac{d}{dz}\,{}_1F_1(a;b;z) = \frac{a}{b}\,{}_1F_1(a+1;b+1;z) \tag{4.3}$$

at $(a,b) = (\alpha,1)$, $z = -\kappa t$.

Proof. By (3.4) the left-hand side of (4.2) is $\partial_{t\,1}F_1(\alpha;1;-\kappa t)$. Size-biasing $\mathrm{Beta}(\alpha, 1-\alpha)$ gives $\mathrm{Beta}(\alpha+1, 1-\alpha)$, whose transform is ${}_1F_1(\alpha+1;2;\cdot)$, with normalizing constant $B(\alpha+1,1-\alpha)/B(\alpha,1-\alpha) = \alpha$. Since $Gy = -\kappa y$, the right-hand side is $-\kappa\alpha_1F_1(\alpha+1;2;-\kappa t)$, and (4.2) becomes (4.3). □

We state this to settle what is and is not new. Identity (4.3) is textbook (Olver *et al.* 2010, Eq. 13.3.15), so Theorem 4.1 carries no new information when $k = 1$. Nor does it when $k > 1$ with *independent* clocks: then μ_α is a product, $\mu_\alpha^{(j)}$ tilts only coordinate j, and (4.2) decomposes into k separate copies of Proposition 4.2. The theorem has content precisely when $k > 1$ and the clocks are dependent, because size-biasing one coordinate of a non-product μ_α perturbs all k marginals of the tilted measure at once.

4.2. *Failure of tensorization*

Proposition 4.3. *Under Assumption 3.1 with $k \geq 2$ and $a_j > 0$, the mixing measure μ_α is not a product measure. Consequently the evolution equation obtained by replacing μ_α with the product of its marginals – the tensorization of the one-clock equation – is incorrect.*

Table 4. The local evolution equation (4.2) and its tensorization (4.4), at $\alpha = 1/2$, $a_1 = a_2 = 1$, $t = 1$. $M'(t)$ is estimated by a central difference on common random paths; "size-biased" is the right-hand side of (4.2).

κ_1	κ_2	$M'(t)$	size-biased	ratio	tensorized	ratio
1	1	-0.29467	-0.29188	0.9905	-0.31517	1.070
0.5	2	-0.26618	-0.26684	1.0025	-0.28464	1.069
2	2	-0.21689	-0.21718	1.0013	-0.24019	1.107
0.3	3	-0.22355	-0.22563	1.0093	-0.23680	1.059
1	0.25	-0.27931	-0.27270	0.9763	-0.28342	1.015

Non-product-ness follows from Proposition 3.4: a product measure with continuous marginals gives the diagonal zero mass, and μ_α does not. The quantitative content is in the last column of Table 4.2. Because the marginals are exactly $\mathrm{Beta}(\alpha, 1-\alpha)$, the tensorized prediction is entirely classical,

$$-\sum_j \kappa_j\, \alpha\, {}_1F_1(\alpha+1; 2; -\kappa_j t) \prod_{i\neq j} {}_1F_1(\alpha; 1; -\kappa_i t), \tag{4.4}$$

and it overstates the true derivative by between 1.015 and 1.107. The common shock is therefore not a correction one can absorb into marginal calibration, even though Proposition 3.2 says the marginals themselves are untouched by it.

5. The Correlation Term Structure Carries No Information

Specialize to the two-component form standard in the staleness literature: an idiosyncratic and a common component, each on its own clock,

$$P_i(t) = \sigma_i\, Y_i\big(H_i(t)\big) \;+\; a_i\, Y_0\big(H_0(t)\big), \tag{5.1}$$

with H_i the undershoot of a standard α_i-stable subordinator, H_0 that of a standard α_c-stable subordinator, and $Y_0, Y_1, \ldots$ independent factors (2.4) with $\sigma_0 = 1$. Write $\rho_{ih}(\Delta)$ for the correlation of the calendar-time returns over $[t, t+\Delta]$.

5.1. *Under staleness alone the term structure is exactly flat*

Proposition 5.1. *Let the factors be driftless Brownian motions, $\kappa_j = 0$. Then for every $t > 0$ and every $\Delta > 0$,*

$$\rho_{ih}(\Delta) \;=\; \frac{a_i a_h\, \alpha_c}{\sqrt{\big(\sigma_i^2\alpha_i + a_i^2\alpha_c\big)\big(\sigma_h^2\alpha_h + a_h^2\alpha_c\big)}}\,, \tag{5.2}$$

exactly. In particular ρ_{ih} does not depend on the return horizon.

The proof is two lines and rests on one exact fact: $\mathbb{E}[H(t)] = \alpha t$, because $H(t)/t \sim \mathrm{Beta}(\alpha, 1-\alpha)$ has mean α. Conditionally on the clocks the Brownian increments have variance $\sigma_i^2\Delta H_i + a_i^2\Delta H_0$ and covariance $a_i a_h \Delta H_0$; taking expectations replaces each ΔH_j by $\alpha_j\Delta$, and Δ cancels. Table 5.1 confirms this end to

Table 5. Proposition 5.1 tested end to end on simulated prices with Brownian factors, $\alpha = 1/2$.

Case	Δ	simulated ρ	s.e.	Eq. (5.2)
1	0.02	0.51876	0.01709	0.50000
1	0.1	0.50334	0.00917	0.50000
1	0.5	0.49764	0.00549	0.50000
1	2	0.49747	0.00435	0.50000
2	0.02	0.22953	0.01128	0.21693
2	0.1	0.21426	0.00645	0.21693
2	0.5	0.22012	0.00527	0.21693
2	2	0.21777	0.00491	0.21693
3	0.02	0.41615	0.01555	0.40000
3	0.1	0.40250	0.00841	0.40000
3	0.5	0.40437	0.00525	0.40000
3	2	0.39480	0.00521	0.40000

end on simulated prices: across 3 parameter sets and a 100-fold range of horizons the largest discrepancy is 1.12 standard errors, and the largest spread of ρ across horizons is 0.02136.

The economic reading is that *staleness by itself generates no correlation term structure at all.* Whatever term structure a calibrated model displays comes from the factor dynamics, not from the staleness mechanism.

Remark 5.1 (why this model has no Epps effect, and what would give it one). Measured correlations between equities rise with the return horizon (Epps 1979), a stylized fact usually attributed to nonsynchronous trading (Lo & MacKinlay 1990). Proposition 5.1 says the specification (5.1) cannot reproduce it, and the reason is structural rather than parametric: in (5.1) both assets observe the common factor through the *same* clock H_0, so their views of it are never out of step and no amount of staleness desynchronizes them. The asynchronicity that generates the Epps effect requires the common factor to be sampled at *asset-specific* times, i.e. $a_i Y_0(H_i(t))$ rather than $a_i Y_0(H_0(t))$. Under that variant the covariance becomes the expected overlap of the intervals $[H_i(t), H_i(t+\Delta)]$ and $[H_h(t), H_h(t+\Delta)]$, which grows with Δ and so does produce a rising term structure – and whose growth rate is governed by how strongly the two clocks are coupled. The distinction matters for identification: it is the second variant, not the one analyzed here, in which the common-shock loading leaves a signature in second moments. We have not developed that case, and flag it as the natural companion to Question 7.1.

5.2. *With mean reversion, the closed form has a domain*

For Ornstein–Uhlenbeck factors, $\mathrm{Var}(Y(u) - Y(v)) = (\sigma^2/\kappa)(1 - e^{-\kappa|u-v|})$, so (5.2) holds with $\alpha_j \Delta$ replaced by $\kappa_j^{-1}\mathbb{E}[1 - e^{-\kappa_j \Delta H_j}]$.

It is worth being explicit about a trap here. One might expect the linearization $\mathbb{E}[1 - e^{-\kappa \Delta H}] \approx \kappa\,\mathbb{E}[\Delta H] = \kappa\alpha\Delta$ to become exact as $\Delta \downarrow 0$, which would make

(5.2) the universal short-horizon limit. It does not. The undershoot increment ΔH is zero on most paths and of order t on the rest – when a trade finally arrives after a long stale period, the clock jumps by the whole staleness – so the relevant scale is κt, not $\kappa\Delta$, and the approximation error does not vanish with Δ. Table 5.2 shows the ratio: for $\kappa t \leq 0.1$ it stays above 0.907, so (5.2) is accurate; for $\kappa t \geq 10$ it never exceeds 0.320, and (5.2) is not usable.

5.3. *The shape of the term structure is not identified*

An earlier conjecture in this project was that common-shock staleness *predicts* the shape of $\Delta \mapsto \rho_{ih}(\Delta)$ – specifically that it forces the correlation to rise. It does not.

We evaluated (5.1) exactly on 139,968 parameter combinations spanning the admissible ranges of $(\alpha_i, \alpha_h, \alpha_c, \sigma_i, \sigma_h, a_i, a_h, \kappa_i, \kappa_h, \kappa_0)$ over 12 horizons from 0.01 to 2, and classified each term structure by its monotonicity. Table 5.3 and Figure 4 report the outcome: every class occurs with substantial frequency and none dominates. The correlation term structure is free, and no empirical term-structure pattern can confirm or refute common-shock staleness on its own.

A weaker claim also fails. We had conjectured that the sign of the initial slope is set by comparing the calendar-time mean-reversion rates $\alpha_j\kappa_j$ and $\alpha_c\kappa_0$. Across the same sweep that rule classifies 52.6% of cells correctly, which is chance. We record it as refuted.

6. Testing the Load-Bearing Assumption

6.1. *Why $\alpha < 1$ is load-bearing*

Every exact statement in Sections 3–5 uses $\alpha \in (0, 1)$, and not incidentally. If the inter-trade duration distribution has a finite mean, then $H(t)/t \to 1$ almost surely by the elementary renewal theorem, the mixing measure μ_α degenerates to a point mass at $(1, \ldots, 1)$, the Beta law (3.2) disappears, (4.1) collapses to $q(t, \cdot) = \prod_j p_j(t, \cdot)$, and the process is Markov again. Staleness does not vanish – prices

Table 6. $\mathbb{E}[1 - e^{-\kappa\Delta H}]/(\kappa\,\mathbb{E}[\Delta H])$ at $\alpha = 1/2$, $t = 1$. A value of one means (5.2) applies. The ratio is governed by κt and does not approach one as $\Delta \downarrow 0$.

κt	$\Delta = 0.02$	$\Delta = 0.1$	$\Delta = 0.5$	$\Delta = 2$
0.03	0.992	0.991	0.987	0.971
0.1	0.974	0.972	0.958	0.907
0.3	0.927	0.920	0.882	0.756
1	0.791	0.775	0.680	0.450
3	0.568	0.536	0.392	0.188
10	0.320	0.272	0.147	0.060
30	0.171	0.117	0.051	0.020
100	0.073	0.038	0.016	0.006

Table 7. Shape of $\Delta \mapsto \rho_{ih}(\Delta)$ across 139,968 admissible parameter combinations of model (5.1).

Term-structure shape	Share of cells (%)
Rising	21.1
Falling	22.6
Hump	31.4
Dip	24.2
Flat	0.7

Note: Grid ranges and the classification rule are in Appendix B.

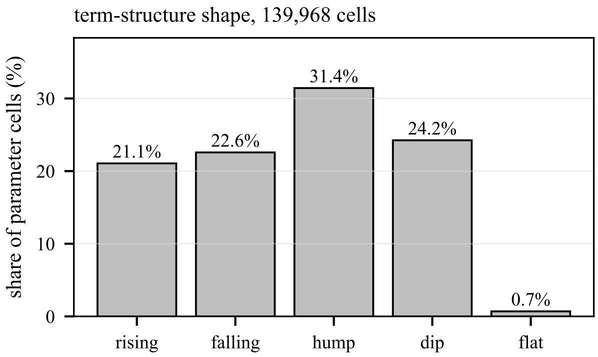

Fig. 4. Monotonicity class of the correlation term structure across 139,968 parameter combinations. All four classes occur with comparable frequency, so the shape carries no identifying information about the staleness mechanism.

are still frozen between trades – but it stops being a non-negligible fraction of any horizon, and the non-Markovian apparatus becomes an asymptotically empty elaboration.

Concretely the model requires $\mathbb{P}[\tau > x] \sim Cx^{-\alpha}$ with $\alpha < 1$ for the inter-trade duration τ: an infinite mean waiting time. This is the assumption the call for papers cites as the empirical motivation for the program. It is testable and the tests are cheap.

6.2. *Data*

- **Cryptocurrency.** Binance spot aggregated-trade records for 21 USDT pairs over one full UTC day, spanning 5 orders of magnitude in 24-hour turnover (from 4,098 to 469,369,949 USDT): 1,332,610 orders built from 3,732,027 individual fills, timestamped to the millisecond. Each record carries the first and last raw trade identifiers, so the number of fills behind every order is known exactly and both event definitions are available from one download.

Table 8. Duration tail index by event definition, Binance spot, 21 pairs. Every column uses the same underlying trade records.

Event definition	median $\hat{\alpha}$	share $\hat{\alpha} < 1$ (%)	pairs
individual fills	0.24	76	21
orders (aggregated fills)	1.19	38	21
orders merged within 0.1 s	1.51	10	21
orders merged within 1 s	1.61	5	21
orders merged within 5 s	2.42	5	21
orders, diurnally adjusted	1.28	35	20

- **Vietnamese equities.** Trade-by-trade records from the Ho Chi Minh Stock Exchange for 30 symbols over one continuous morning session, 28,696 prints in total, with median inter-trade gaps ranging from 1 to 44 seconds. HOSE timestamps have one-second resolution, and between 0% and 20% of consecutive prints tie.

The two markets are deliberately opposite: a continuously traded twenty-four-hour venue with millisecond stamps, and a session-based equity market with second stamps. Estimation procedures – the Hill estimator (Hill 1975) with a Hill-plot plateau rule, an assumption-free running-mean diagnostic, burst aggregation, diurnal adjustment, and the Clauset–Shalizi–Newman goodness-of-fit test (Clauset *et al.* 2009) – are in Appendix B.

6.3. *The tail index is a property of the event definition*

A marketable order executed against several resting orders generates several prints at one timestamp. Counting each fill as an event injects zero durations, which enlarges the sample and pushes the Hill threshold deep into the body of the distribution. Table 6.3 recomputes the tail index under successively coarser event definitions on the identical data.

The median index rises monotonically from 0.24 at fill level to 1.19 at order level and 1.61 after merging within one second, and the share of pairs with $\hat{\alpha} < 1$ falls from 76% to 38% to 5%. Figure 5 shows the whole distribution.

This is diagnostic, not merely inconvenient. A tail index is an asymptotic property of the duration distribution and is invariant under coarsening of the event definition; a statistic that moves monotonically and without bound as the definition is coarsened is measuring the microstructure of order execution, not the tail of a trading-time distribution. The apparent infinite-mean waiting times in the raw data are an artifact of the finest event definition.

They are also largest where markets are most active, which is worth stating because it inverts the intuition that thin trading is what produces heavy duration tails. Fills per order range from 1.02 to 3.80 across the sample and rise steeply with turnover (correlation with log 24-hour volume +0.818, $p < 0.001$); the share of fill-level durations that are exactly zero runs from 14% to 81% with median 60%, and it too rises with volume (+0.837, $p < 0.001$). It is the liquid pairs that burst,

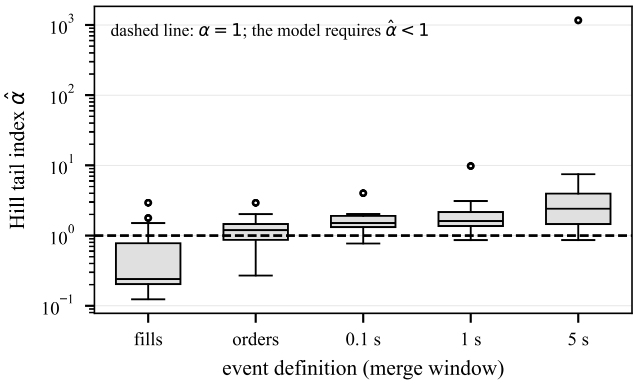

Fig. 5. Distribution across 21 Binance pairs of the Hill tail index under successively coarser event definitions, from individual fills through orders to orders merged within a window. The model requires the index to lie below the dashed line. A genuine tail index would be invariant to this choice.

and therefore the liquid pairs whose raw tail index is most badly under-estimated.

6.4. *The diurnal cycle does the rest*

A Poisson process whose rate varies over the day produces durations that are a *mixture* of exponentials, and mixtures of exponentials have heavier-than-exponential tails. That a simple mechanism of this kind can manufacture an apparent power law is well established for returns (LeBaron 2001, Farmer & Lillo 2004); the point here is that inter-trade durations are exposed to it just as directly, and over a trading day the modulating rate is not a subtle effect. Any index estimated on a full trading day therefore confounds tail behavior with the activity cycle. The cycle is large: the smoothed intraday profile of mean duration varies by a median factor of 8.4 across pairs, and by up to 196.

Dividing durations by that profile changes two things. The assumption-free running-mean ratio – which drifts upward without bound when the mean is infinite – tightens from a raw range of 0.42–3.79 to 0.74–1.89, so the apparent non-convergence was the activity cycle rather than an infinite mean. And the median tail index rises to 1.28, with 35% of pairs below one. Figure 6 traces each symbol individually through the adjustment.

6.5. *The decisive test: is anything both a power law and infinite-mean?*

The Hill estimator returns a number whether or not the tail is a power law, and it uses a threshold chosen by rule rather than fitted. The Clauset–Shalizi–Newman procedure removes both objections: it fits the threshold $x_{\min}$ and the index by

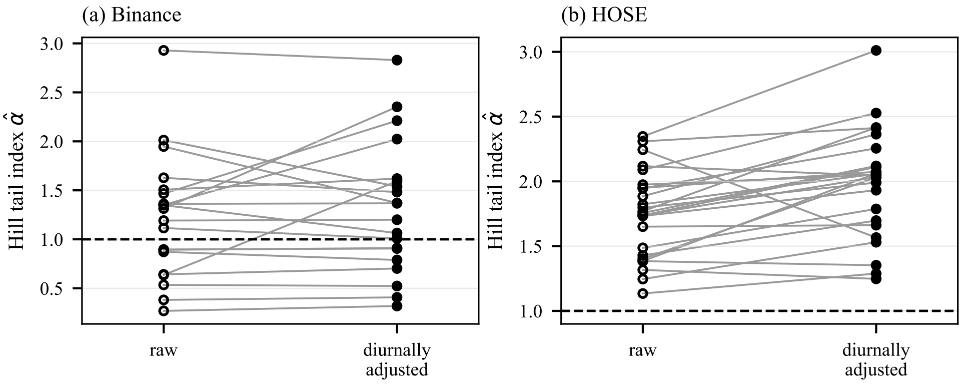

Fig. 6. Effect of removing the diurnal activity cycle on the estimated tail index, symbol by symbol, in both markets. The dashed line is $\alpha = 1$, below which the model requires every point to lie.

maximum likelihood and calibrates the Kolmogorov–Smirnov distance against parametric bootstrap replications, so a small p-value rejects the power-law hypothesis itself.

Applying it to every series in both markets gives the paper's sharpest result. Among the 21 cryptocurrency pairs, the number that are both a plausible power law ($p \geq 0.10$) and infinite-mean ($\hat{\alpha} < 1$) is 0; after diurnal adjustment it is 0. Among the 29 HOSE symbols it is 0. The fitted indices are not marginal: their median is 1.73 at order level and 2.55 after adjustment, with a minimum across all adjusted series of 1.13.

Two points of interpretation. First, the fitted threshold $x_{\min}$ always falls above the zero durations, so the positive-duration sample is the same under the fill- and order-level definitions and the test is properly a statement about the order-level series; the fill-level zeros sit in the body below $x_{\min}$ by construction. That is precisely why the Hill estimator and this procedure disagree so sharply: Hill's threshold is a fixed fraction of n and the zeros inflate n, whereas a fitted $x_{\min}$ is immune to them. Second, some series do have a fitted index below one – the smallest at order level is 0.29 – but in every such case the power-law hypothesis is itself rejected, so the index estimates the tail exponent of a distribution that has none.

The pairs that appeared to support $\alpha < 1$ under the Hill estimator are precisely those for which a properly fitted threshold puts the index above one, or for which the power law is rejected outright. Per-symbol detail is in Tables 6.6 and 6.6.

6.6. *Cross-section: illiquid names do not have heavier tails*

If staleness were a tail phenomenon, illiquid names would have *smaller* indices: their durations would be not merely longer but heavier-tailed. The model's cross-sectional prediction is a strong negative correlation between typical gap length and $\hat{\alpha}$. We find the opposite sign in both markets.

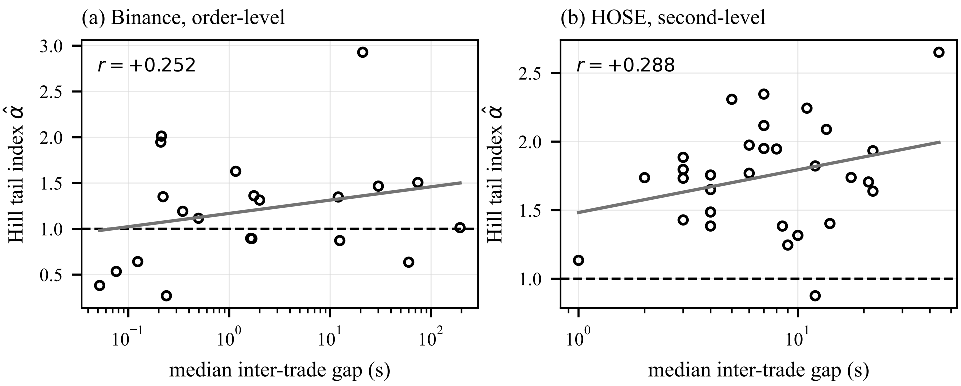

Fig. 7. Tail index against median inter-trade gap. The model requires all points below the dashed line and a steeply negative slope; the fitted slope is positive in both markets.

On Binance the correlation between the log median gap and the tail index is +0.252 ($p = 0.270$) at order level and +0.484 ($p = 0.031$) after diurnal adjustment. On HOSE it is +0.288 ($p = 0.123$, $n = 30$) and -0.016 ($p = 0.941$) respectively. Figure 7 shows both cross-sections. Illiquid names have longer typical gaps and the same, or slightly lighter, tails. Staleness in these markets is a scale effect, not a tail effect.

6.7. *Testing the prediction that does survive*

Everything so far has tested the tail assumption. It is worth asking separately whether the framework's *dependence* prediction holds, because Section 7 argues that this is the part robust to the tail index, and because Propositions 3.2 and 5.1 have already ruled out the two easier places to look: marginal durations are blind to the common shock, and the correlation term structure is flat.

What remains is the atom itself. Bin calendar time at width b and let $A_i(k)$ indicate that asset i traded in bin k. Independent clocks make A_i and A_h independent; a shared clock component makes them positively associated. The 2×2 table of co-trading counts and its odds ratio are therefore the natural identifying moment, and unlike the atom itself they are directly observable.

Two things would otherwise fake a positive result. Both assets are quiet at the same times of day, so we stratify by time-of-day bucket (30 minutes for Binance, 10 for HOSE) and pool with the Mantel–Haenszel estimator; the unstratified ratio is reported alongside so the size of that confound is visible. And bins are strongly autocorrelated, so confidence intervals come from a block bootstrap over whole strata, 400 replications.

Table 6.7 reports the result, and it is a qualified confirmation. The sign is right and it survives the control: at the finest resolution the median Mantel–Haenszel odds ratio is 1.03 on Binance (91 pairs, 80% above one, 32% individually significant) and

1.13 on HOSE (435 pairs, 76% above one, 27% significant). Stratification moves the estimate only from 1.08 to 1.03 and from 1.16 to 1.13, so the diurnal cycle accounts for little of it. Trade arrival is genuinely dependent across assets beyond the common activity cycle, as a shared clock component requires.

The magnitude, however, points the same way as the rest of this section. To read it we need the model's own value of the same statistic, computed by generating the trade instants as the range of the clock and binning them identically (Table 6.7, 4,000 paths). The instructive entry is the first row. With *no* common shock at all, $a = 0$, the model still produces an odds ratio of 1.37–1.87, not one: an α-stable clock is bursty at every time scale, so activity clustering survives any finite stratification. That is the correct null for this model, and the observed values of 1.03 and 1.13 lie *below* it.

So the framework is not refuted here, but it is over-strong in a second, independent way. Real trade arrivals are positively dependent across assets, as the common-shock construction predicts; they are simply far less clustered than inverse-stable subordination makes them even before any common shock is added. Within this model class the implied loading is therefore indistinguishable from zero, not because the dependence is absent but because the baseline is already too high. Recovering a usable estimate of a_j needs the finite-mean clock of Section 7, whose burstiness is controlled by the tempering rate; we state that as part of Question 7.1.

6.8. *Summary*

1. The estimated duration tail index rises monotonically as the event definition is coarsened, from 0.24 at fill level to 2.42 at a five-second merge. A genuine tail index would not move.
2. Removing the diurnal activity cycle raises it further, to 1.28, and removes the apparent non-convergence of running means.
3. Once the tail threshold is fitted rather than chosen by rule, no series in either market – 21 cryptocurrency pairs and 29 equities – is both a plausible power law and infinite-mean.
4. Cross-sectionally the index rises, not falls, with illiquidity, in both markets.
5. The framework's one surviving prediction – that assets sharing a clock component trade and freeze together – does hold in sign and survives the diurnal control, but the observed dependence is weaker than an α-stable clock generates with no common shock at all.

We therefore find no empirical anchor for the inverse-stable-subordinator staleness mechanism as it is usually motivated.

7. Discussion: Staleness as a Scale Effect

7.1. *What survives*

The theory splits cleanly along the line the data draw.

Survives. Everything that follows from *dependence* rather than from *scaling.* Proposition 3.4 and Theorem 3.1 use only that the clocks share jumps, and hold for any multivariate subordinator with a common-shock component whatever the tail index. So does Proposition 4.3: marginal operators never tensorize in the presence of a common shock, because the diagonal atom is invisible to each marginal separately. Combined with Proposition 3.2 – marginals are blind to the common shock – this identifies the atom as the object worth estimating, and it is the part of this paper we expect to be reusable.

Section 6.7 puts that to the test and returns a qualified answer. The dependence is present and survives the diurnal control, so the mechanism is not refuted. But it is weaker than the α-stable clock delivers with no common shock at all, which is the same verdict the tail analysis reaches by an entirely different route: the construction is not wrong about *what* couples two stale assets, only about *how violently* it does so. That is a statement about the clock's marginal law, and it is repaired by the same substitution as the tail problem.

Does not survive. Everything that uses $\alpha \in (0,1)$: the exact Beta mixing law, the self-similarity of Proposition 3.1, the mixture representation (4.1), and hence Theorem 4.1, which is derived from the scaling and not from the time change (Remark 4.1). Proposition 5.1 likewise uses $\mathbb{E}[H(t)] = \alpha t$.

7.2. *The repair the data suggest*

The HOSE cross-section is not merely a rejection; it has a shape. Median gaps span 1 to 44 seconds while indices stay in a band around 1.75 with no downward trend. Illiquidity moves the *scale* of the duration distribution and leaves its *tail* alone.

The clock family that reproduces this is the tempered stable subordinator (Rosiński 2007). Replace each X_j by a subordinator with Lévy measure

$$\Pi_j(dz) = \frac{\alpha}{\Gamma(1-\alpha)}\, z^{-1-\alpha}\, e^{-\theta_j z}\, dz, \qquad \theta_j > 0, \tag{7.1}$$

so that

$$\psi_j(\lambda) = (\lambda + \theta_j)^\alpha - \theta_j^\alpha, \qquad \mathbb{E}[X_j(1)] = \psi_j'(0) = \alpha\, \theta_j^{\alpha-1} < \infty. \tag{7.2}$$

Three properties match the data point for point.

1. **A common index and an asset-specific scale.** All assets share α; heterogeneity enters only through θ_j, which sets the duration scale $1/\theta_j$. That is the observed cross-section.
2. **Finite mean.** By (7.2) the expected duration is finite for every $\theta_j > 0$, consistent with the running-mean ratios of Section 6 once the diurnal cycle is removed.
3. **A crossover, not an exponent.** For $t \ll 1/\theta_j$ the tempering is invisible and $H_j(t)/t$ is approximately $\mathrm{Beta}(\alpha, 1-\alpha)$: at intraday horizons the process behaves exactly like the stable clock analyzed above. For $t \gg 1/\theta_j$, $H_j(t)/t \to 1$ and the process is effectively Markov. Staleness becomes a phenomenon of a specific estimable time scale rather than a self-similar fraction of every horizon.

The common-shock construction (2.3) carries over unchanged, so the simultaneity atom and Theorem 3.1 continue to hold with the tempered Lévy tail in place of (2.2). What is lost is exact self-similarity, and with it Theorem 4.1: the mixing measure then depends on t, and the local equation survives only in the short-horizon regime.

Question 7.1. For a multivariate tempered stable clock $S_j = X_j + a_j Z$ with common index α and asset-specific tempering rates θ_j, what governing equation does the law of $(Y_1(H_1(t)), \ldots, Y_k(H_k(t)))$ satisfy, and does the crossover at $t \sim 1/\theta_j$ admit a uniform description?

7.3. *Implications for the non-Markovian factor program*

Three conclusions for the wider program.

First, the two strands the program seeks to unify may not need the same mathematics. Long memory in volatility is empirically robust (Gatheral *et al.* 2018). Infinite-mean inter-trade durations are not: on the evidence here they are an artifact of fill-level data and the diurnal cycle. A framework coupling the two through a shared index α imposes a constraint the data do not support.

Second, staleness itself is not in question. Prices do freeze between trades and excess idle time is a real, measurable feature of equity data (Bandi *et al.* 2017). What is in question is the specific mechanism of an α-stable trading clock with $\alpha < 1$. Non-exponential durations with finite mean – Weibull, log-normal, tempered stable, or the self-exciting arrival models of Bacry *et al.* (2013) and Engle & Russell (1998) – are compatible with everything we observe and produce staleness without infinite mean.

Third, the multivariate structure is where the mathematical interest remains. The simultaneity atom is invisible to any single asset's duration distribution (Proposition 3.2) and not recoverable from the correlation term structure (Proposition 5.1 and Table 5.3); it can be identified only from joint staleness patterns. Theorem 3.1 makes it computable. That is a well-posed estimation problem, unaffected by our negative tail result, and it does not appear to have been attempted.

8. Conclusion

We have built the multivariate inverse-subordination factor model that the additive-subordination literature identified as its open next step and characterized it: exact self-similarity of the clock vector, generalized arcsine marginals that are blind to the common shock, an atom of simultaneous staleness with a compensation-formula representation, a closed-form bivariate potential density at $\alpha = 1/2$ whose logarithmic diagonal ridge explains a numerical discrepancy previously left open, an exact local evolution equation driven by size-biased mixing measures that reduces to a classical confluent hypergeometric identity for one clock, and a quantified failure of the marginal operators to tensorize.

We have also reported negative results a reader needs in order to use any of this. Under staleness alone the correlation term structure is exactly flat, and once mean reversion is added its shape is unidentified: all four monotonicity classes occur with comparable frequency across 139,968 parameter combinations, and a slope-sign rule we had conjectured performs at chance. Most importantly, the mechanism's load-bearing assumption – a duration tail index below one – has no empirical support in either market we tested. The estimated index is a property of the event definition and the diurnal cycle, and once the tail threshold is fitted properly no series is both a plausible power law and infinite-mean.

We also tested the one prediction that does not rest on the tail index – that assets sharing a clock component trade together. It holds in sign, and survives stratifying out the diurnal cycle, but it is weaker in the data than the model produces with the common shock switched off entirely.

The constructive reading is that staleness in modern electronic markets is a scale effect rather than a tail effect, and that a tempered stable clock with a common index and asset-specific tempering rates is the model the data describe. The dependence results of this paper transfer to that setting unchanged, and the tempering rate is what would let the common-shock loading finally be estimated rather than merely bounded; the scaling results do not transfer, and finding the governing equations for the tempered multivariate case is the problem we would work on next.

Appendix A. Proofs

Proof of Proposition 3.1

A standard α-stable subordinator satisfies $(X(c^\alpha u))_{u\geq 0} \overset{d}{=} (c\,X(u))_{u\geq 0}$. Under Assumption 3.1 all of $X_1, \dots, X_k, Z$ share the index and are independent, so the scaling holds jointly for the vector, hence for $S = (S_1, \dots, S_k)$. Substituting $u = c^\alpha v$,

$$L_j(ct) = \inf\{u : S_j(u) > ct\} = c^\alpha \inf\{v : S_j(c^\alpha v) > ct\} \overset{d}{=} c^\alpha \inf\{v : c\,S_j(v) > ct\} = c^\alpha L_j(t), \tag{A.1}$$

jointly in j, since one realization of the scaled process serves every coordinate. Hence $H_j(ct) = S_j(L_j(ct)-) \overset{d}{=} c\,S_j(L_j(t)-) = c\,H_j(t)$, jointly. Taking $c = 1/t$ gives the last claim. □

Proof of Proposition 3.3

By Proposition 3.2, $H(t) = tB$ with $B \sim \mathrm{Beta}(\alpha, 1-\alpha)$. The moment generating function of $\mathrm{Beta}(a, b)$ is ${}_1F_1(a; a+b; \cdot)$ (Olver *et al.* 2010, Sec. 13); with $a = \alpha$, $b = 1-\alpha$ we have $a + b = 1$, giving (3.3). For (3.4), condition on $H_j(t)$, which is independent of Y_j, and use $\mathbb{E}[Y(u)|Y(0) = y] = ye^{-\kappa u}$. □

Proof of Proposition 3.4 and Theorem 3.1

Since X_j and X_h are independent stable subordinators they a.s. share no jump time, so S_j and S_h jump simultaneously precisely at the jump times of Z. If $L_j(t) =$

$L_h(t) = u$, both coordinates cross t at u, forcing $\Delta Z(u) > 0$. Conversely, at a jump time u of Z with pre-jump position (s_j, s_h) and jump size ζ, both cross t iff $s_j, s_h \le t$ and

$$\zeta > \frac{t - s_j}{a_j} \vee \frac{t - s_h}{a_h}. \tag{A.2}$$

The jump times and sizes of Z form a Poisson point process with intensity $du\,\Pi_{\alpha_c}(d\zeta)$; the pre-jump position is $\mathcal{F}_{u-}$-measurable and the jump size independent of it, so the compensation formula gives

$$\begin{aligned}\mathbb{P}\big[L_j(t) = L_h(t)\big] &= \int_0^\infty \mathbb{E}\left[\mathbf{1}\{S_j(u) \le t,\, S_h(u) \le t\}\, \bar{\Pi}_{\alpha_c}\Big(\tfrac{t-S_j(u)}{a_j} \vee \tfrac{t-S_h(u)}{a_h}\Big)\right] du \\ &= \int_{[0,t]^2} \bar{\Pi}_{\alpha_c}\Big(\tfrac{t-s_j}{a_j} \vee \tfrac{t-s_h}{a_h}\Big)\, U_{jh}(ds_j, ds_h),\end{aligned} \tag{A.3}$$

which is (3.6). Positivity holds because $\bar{\Pi}_{\alpha_c} > 0$ on $(0, \infty)$ by (2.2) and U_{jh} charges $[0, t]^2$; $\pi_{jh} < 1$ because with positive probability X_j alone carries S_j across t strictly before S_h crosses. Independence of t follows from Proposition 3.1, the event being invariant under the joint scaling (3.1). If $a_j a_h = 0$, at least one coordinate shares no jump with the other and simultaneous crossing has probability zero. □

Proof of Proposition 3.5

For Borel $B \subset (0, \infty)^2$ and $c > 0$, substituting $v = c^\alpha w$ and using the joint scaling above,

$$U_{jh}(cB) = \int_0^\infty \mathbb{P}\big[(S_j, S_h)(v) \in cB\big]\, dv = c^\alpha \int_0^\infty \mathbb{P}\big[c\,(S_j, S_h)(w) \in cB\big]\, dw = c^\alpha\, U_{jh}(B). \tag{A.4}$$

Applying this to a small box of measure $|B|$, so $|cB| = c^2|B|$, and letting $|B| \to 0$ gives (3.7) at every Lebesgue point. □

Proof of Theorem 3.2

The standard $\frac{1}{2}$-stable subordinator, with Laplace exponent $\lambda^{1/2}$, has density

$$f_u(x) = \frac{u}{2\sqrt{\pi}}\, x^{-3/2}\, e^{-u^2/(4x)}, \qquad x > 0, \tag{A.5}$$

as follows from $\int_0^\infty e^{-\lambda x} f_u(x)dx = e^{-u\sqrt{\lambda}}$. Conditioning on the common component and using independence, the joint density of $(S_j(u), S_h(u))$ at (s_j, s_h) is $\int_0^{(s_j \wedge s_h)/a} f_u(x_j) f_u(x_h) f_u(z)\, dz$ with $x_j = s_j - az$, $x_h = s_h - az$; the substitution $w = az$ for the scaled shock contributes no net Jacobian, and the upper limit enforces $x_j, x_h > 0$. Integrating over operational time and exchanging the order of integration,

$$u(s_j, s_h) = \int_0^{(s_j \wedge s_h)/a} \left[\int_0^\infty f_u(x_j) f_u(x_h) f_u(z)\, du\right] dz. \tag{A.6}$$

Inserting (A.5) three times, the inner integral is

$$\frac{(z\,x_j\,x_h)^{-3/2}}{8\pi^{3/2}}\int_0^\infty u^3 e^{-u^2 C/4}\,du, \qquad C := \frac{1}{z}+\frac{1}{x_j}+\frac{1}{x_h}. \tag{A.7}$$

Since $\int_0^\infty u^3 e^{-\lambda u^2}\,du = 1/(2\lambda^2)$, taking $\lambda = C/4$ gives $8/C^2$, so the inner integral equals $\pi^{-3/2}(zx_jx_h)^{-3/2}C^{-2}$. Writing $C = (x_jx_h + z(x_j+x_h))/(zx_jx_h)$ turns this into $\pi^{-3/2}\sqrt{zx_jx_h}\,(x_jx_h + z(x_j+x_h))^{-2}$; integrating in z gives (3.8). $\square$

Proof of Corollary 3.1

Set $s_j = s$, $s_h = s+\delta$, and let $\epsilon := x_j = s - az$, so $x_h = \epsilon+\delta$ and $dz = -d\epsilon/a$. As $\epsilon \downarrow 0$ we have $z \to s/a$, and $x_jx_h = O(\epsilon^2+\epsilon\delta)$ is negligible against $z(x_j+x_h) \asymp (s/a)(2\epsilon+\delta)$, so the integrand of (3.8) behaves like

$$\frac{\sqrt{z}\,\sqrt{\epsilon(\epsilon+\delta)}}{z^2(2\epsilon+\delta)^2} \asymp \frac{a^{3/2}}{s^{3/2}}\,\frac{\sqrt{\epsilon(\epsilon+\delta)}}{(2\epsilon+\delta)^2}. \tag{A.8}$$

For $\epsilon \gg \delta$ this is $\asymp 1/(4\epsilon)$, so the range $\delta \lesssim \epsilon \lesssim s$ contributes $\asymp \frac{1}{4}a^{1/2}s^{-3/2}\log(s/\delta)$ to $\int d\epsilon/a$, while $\epsilon \lesssim \delta$ contributes $O(1)$. Hence $u(s, s+\delta) = -\pi^{-3/2}c(s)\log\delta + O(1)$ with $c(s) = \frac{1}{4}a^{1/2}s^{-3/2} > 0$. $\square$

Proof of Theorem 4.1

By Proposition 4.1 and dominated convergence,

$$\partial_t q(t,y) = \int_{[0,1]^k}\sum_{j=1}^k s_j\,\big(\partial_r p_j\big)(ts_j, y_j)\prod_{i\neq j} p_i(ts_i, y_i)\ \mu_\alpha(ds). \tag{A.9}$$

Each Y_j is Feller, so $\partial_r p_j = G_j p_j$, and G_j acts only on the jth coordinate, hence commutes with the s-integration:

$$\partial_t q(t,y) = \sum_{j=1}^k G_j \int_{[0,1]^k} s_j \prod_{i=1}^k p_i(ts_i, y_i)\ \mu_\alpha(ds). \tag{A.10}$$

By Proposition 3.2 the jth marginal of μ_α is $\mathrm{Beta}(\alpha, 1-\alpha)$, of mean α, so $s_j\mu_\alpha(ds) = \alpha\mu_\alpha^{(j)}(ds)$ and the integral equals $\alpha q^{(j)}(t,y)$. $\square$

Proof of Proposition 5.1

Let $\Delta H_j := H_j(t+\Delta) - H_j(t) \geq 0$. Conditionally on the clocks, the Brownian increments are independent with $\mathrm{Var}(\Delta P_i \mid \cdot) = \sigma_i^2\Delta H_i + a_i^2\Delta H_0$ and $\mathrm{Cov}(\Delta P_i, \Delta P_h \mid \cdot) = a_ia_h\Delta H_0$. All conditional means vanish, so the unconditional moments are the expectations of these. By Proposition 3.2, $\mathbb{E}[H_j(t)] = \alpha_j t$ exactly, hence $\mathbb{E}[\Delta H_j] = \alpha_j\Delta$. Therefore

$$\rho_{ih}(\Delta) = \frac{a_ia_h\alpha_c\Delta}{\sqrt{(\sigma_i^2\alpha_i + a_i^2\alpha_c)\Delta\ (\sigma_h^2\alpha_h + a_h^2\alpha_c)\Delta}}, \tag{A.11}$$

and Δ cancels, giving (5.2) for every $\Delta > 0$. $\square$

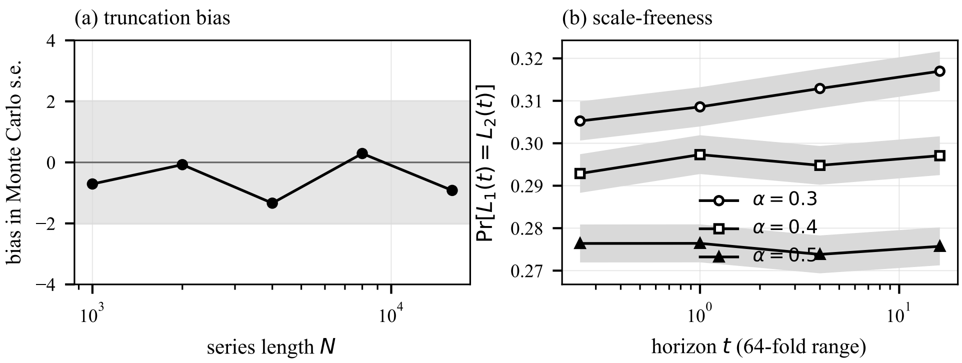

Fig. 8. Left: realized bias of the simulated normalized undershoot against its exact Beta mean, in Monte Carlo standard errors, as a function of the series length; the shaded band is ± 2. Right: the simultaneity atom over a 64-fold range of horizons, with two-standard-error bands, for three values of α.

Appendix B. Simulation and Estimation

Exact simulation of the clock vector

Every simulation in this paper enumerates jumps rather than discretizing time. For a subordinator with Lévy measure Π, the jumps on the operational interval $[0, U]$ are $\{(V_i, \bar{\Pi}^{-1}(\Gamma_i/U))\}_{i\geq 1}$ with Γ_i the arrival times of a unit-rate Poisson process and V_i i.i.d. uniform on $(0, U)$. For the standard α-stable subordinator this gives the ith largest jump as $J_i = (U/(\Gamma_i \Gamma(1-\alpha)))^{1/\alpha}$. Merging the jumps of $X_1, \ldots, X_k$ and Z in operational time and taking cumulative sums yields the exact step path of every S_j at its jump points.

This matters for two claims. Because the path is known exactly at its jumps, $H_j(t)$ and $L_j(t)$ are computed exactly, and the event $\{L_j(t) = L_h(t)\}$ is decided by comparing *event indices* – so the simultaneity atom of Section 3.4 cannot be a discretization artifact. And because S is a step function of u, the occupation integral defining the potential measure is evaluated exactly rather than approximated.

Truncating the series after N terms omits an expected mass $(U/\Gamma(1-\alpha))^{1/\alpha}\frac{\alpha}{1-\alpha}N^{-(1-\alpha)/\alpha}$ over $[0, U]$. That bound is very conservative, because what matters is the mass omitted over $[0, L(t)]$ rather than over $[0, U]$. We therefore selected N from measured bias rather than from the bound: with $N = 8{,}000$ the bound is 2.5×10^{-3}, while the realized bias in $\mathbb{E}[H(t)/t]$ against its exact value never exceeds 1.3 Monte Carlo standard errors and the smallest Kolmogorov–Smirnov p-value in the convergence sweep is 0.06 (Figure 8, left).

Remark Appendix B.1 (range of α covered). The number of series terms needed for a given accuracy grows like $N \sim \varepsilon^{-\alpha/(1-\alpha)}$, so the exact simulator becomes expensive as $\alpha \uparrow 1$: at $\alpha = 0.7$ and the horizons used here the conservative bound already calls for order 10^9 terms. The simulation evidence in this paper is therefore confined to $\alpha \in \{0.3, 0.4, 0.5\}$ for the joint-law tests and to $\alpha = 1/2$ for the atom, potential density and evolution equation, where the closed form of Theorem 3.2 is also available. The correlation sweep reaches $\alpha = 0.65$ because it uses the tabulated increment transform rather than fresh paths. All of the propositions are proved for general $\alpha \in (0, 1)$; what is restricted is the numerical corroboration, and the restriction is one of computational cost rather than of validity.

Remark Appendix B.2 (a residual finite-N effect). At the smallest index studied, $\alpha = 0.3$, the atom estimate drifts mildly upward across horizons at $N =$ 8,000. Repeating the comparison at $N = 32{,}000$ removes it: the difference between the extreme horizons falls from $+1.3$ to -0.4 standard errors. The drift is therefore a truncation effect, not a failure of Proposition 3.4, and it is why the self-similarity p-values in Table 3.1 are weakest at small α.

Testing the evolution equation

With independent Ornstein–Uhlenbeck factors and test function $\varphi(y) = y_1 y_2$, every term of (4.2) reduces to the joint Laplace transform $M(t) = \mathbb{E}[\exp(-\kappa_1 H_1(t) - \kappa_2 H_2(t))]$: the equation becomes $M'(t) = -\mathbb{E}[(\kappa_1 s_1 + \kappa_2 s_2)e^{-t(\kappa\cdot s)}]$ with $s_j = H_j(t)/t$, and the tensorized alternative is (4.4). $M'(t)$ is estimated by a central difference evaluated on the *same* simulated paths at $t \pm h$, which removes almost all Monte Carlo noise from the difference quotient.

The correlation sweep

The sweep of Table 5.3 runs over $\alpha_i, \alpha_h, \alpha_c \in \{0.2, 0.35, 0.5, 0.65\}$, $\sigma_i, \sigma_h \in \{0.5, 1, 2\}$, $a_i, a_h \in \{0.5, 1, 2\}$ and $\kappa_i, \kappa_h, \kappa_0 \in \{0.1, 1, 10\}$, on 12 horizons geometrically spaced from 0.01 to 2. The function $\mathbb{E}[1 - e^{-\kappa\Delta H}]$ is tabulated once by exact simulation on a logarithmic grid of κ and interpolated, so the sweep involves no further simulation. A term structure is classified as rising if its maximum is at the last horizon and its minimum at the first, falling in the mirror case, hump if the maximum is interior, dip if the minimum is interior, and flat if the total variation is below a relative tolerance of 10^{-4}.

Duration estimators

Hill. For order statistics $\tau_{(1)} \geq \cdots \geq \tau_{(n)}$ the Hill estimator at threshold k is $\hat{\alpha}_k = (k^{-1}\sum_{i\leq k}\log(\tau_{(i)}/\tau_{(k+1)}))^{-1}$ (Hill 1975). We report the value at the flattest window of the Hill plot, chosen as the run of nine consecutive k minimizing the standard deviation of $\hat{\alpha}_k$, which removes the arbitrary single-threshold choice. Zero durations are retained: they count toward n and hence toward the choice of k, which is precisely the channel through which bursts bias the estimator.

Running mean. The ratio of the sample mean over the second half of the duration sequence to that over the first. For a finite mean this tends to one; for an infinite mean it grows with sample size. It uses no threshold and no parametric assumption.

Burst aggregation. Given a window W, consecutive events whose timestamps differ by less than W are merged and durations recomputed. On HOSE the analogous operation merges prints that share a one-second timestamp.

Diurnal adjustment. Durations are divided by a smoothed profile of the binwise mean duration over the observation window (48 bins for the twenty-four-hour crypto sample, 24 for the HOSE session), interpolated across empty bins, smoothed by a three-bin moving average and normalized to mean one.

Goodness of fit. Following Clauset *et al.* (2009), $x_{\min}$ is chosen to minimize the Kolmogorov–Smirnov distance, α is fitted by maximum likelihood above it, and the p-value is obtained from parametric bootstrap replications that resample the body non-parametrically and draw the tail from the fitted power law. A small p-value rejects the power law; we treat $p < 0.10$ as rejection.

Appendix C. Reproduction

The accompanying repository contains the clock simulator and closed forms (`src/nmf/`), the collectors for both data sets (`data/`), one script per experiment (`experiments/`), and the code that converts the resulting JSON into the tables, figures and numerical macros of this manuscript. Every experiment is seeded, and every number quoted above is generated rather than transcribed: rebuilding the paper from raw data requires running the collectors, the experiment scripts, and `make`.

The crypto sample is one UTC day and the HOSE sample one morning session; both are archived with the code, since public trade endpoints do not serve deep history. The two limitations we regard as material are stated in the text: the HOSE session is short, and its one-second timestamp resolution caps the finest event definition available in that market.

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Table 9. Binance spot, one UTC day. "fill", "order" and "adj." are Hill indices at fill level, order level and after diurnal adjustment; CSN p is the Clauset–Shalizi–Newman goodness-of-fit p-value for the order-level series.

Pair	24h volume (USDT)	orders	median gap (s)	fills /order	Hill $\hat{\alpha}$ fill	Hill $\hat{\alpha}$ order	Hill $\hat{\alpha}$ adj.	CSN p
BTC	469,369,949	185,781	0.346	2.91	0.24	1.19	1.20	0.00
ETH	200,616,691	107,772	0.496	3.80	0.20	1.12	1.01	0.00
XPL	177,849,163	237,525	0.213	2.23	1.78	2.01	1.54	0.47
ALLO	103,539,172	175,614	0.221	3.21	0.24	1.35	2.02	0.41
PLUME	79,828,878	170,520	0.210	2.02	0.77	1.95	1.37	0.07
SOL	62,565,619	48,469	1.159	2.88	0.17	1.63	1.48	0.00
XRP	35,249,120	29,505	1.757	3.08	0.17	1.36	1.37	0.00
HEMI	23,990,471	118,193	0.124	3.79	0.23	0.64	0.70	0.70
LINK	12,732,843	120,988	0.076	3.25	0.23	0.53	0.52	0.00
BICO	8,441,674	94,197	0.052	1.98	0.23	0.38	0.41	0.00
SPCXB	4,934,827	18,139	1.674	1.83	0.69	0.89	0.91	0.00
TRUMP	3,008,634	7,032	1.630	2.74	0.16	0.90	0.91	0.00
SYRUP	1,754,859	6,168	2.001	2.64	0.15	1.31	2.35	0.00
RSR	235,711	1,944	30.000	1.31	1.48	1.47	2.21	0.00
AMP	142,844	2,135	12.015	1.30	0.71	1.35	1.06	0.09
A	85,770	2,449	0.238	1.69	0.27	0.27	0.32	0.00
LQTY	51,617	1,212	12.440	1.77	0.12	0.87	0.79	0.81
XVS	30,203	3,575	21.009	1.20	2.93	2.93	2.83	0.00
PYPLB	18,279	627	73.871	1.11	1.51	1.51	1.62	0.09
SMHB	10,982	516	60.000	1.03	0.64	0.64	1.59	0.02
GSB	4,098	249	193.519	1.02	1.01	1.01	—	0.04

Table 10. HOSE, one continuous morning session. “print” counts every print, “second” merges prints sharing a timestamp, “adj.” additionally removes the intraday activity profile.

Symbol	prints	ties (%)	median gap (s)	print	Hill $\hat{\alpha}$ second	adj.	CSN p
SHB	2,448	12	1.0	1.12	1.13	1.29	0.00
SSI	2,840	20	2.0	1.45	1.74	2.11	0.06
HPG	2,009	12	3.0	1.44	1.73	1.93	0.00
MSN	1,064	9	3.0	1.28	1.43	1.70	0.00
VIX	2,112	12	3.0	1.55	1.80	1.99	0.00
TCB	2,071	9	3.0	1.92	1.89	2.36	0.00
DXG	1,120	12	4.0	1.40	1.38	2.06	0.05
FPT	1,459	9	4.0	1.91	1.49	1.79	0.00
VNM	1,294	11	4.0	1.66	1.65	1.66	0.00
GEX	1,236	10	4.0	1.75	1.76	2.12	0.00
MBB	1,323	8	5.0	1.90	2.31	2.41	0.03
VCB	956	7	6.0	1.78	1.77	2.42	0.00
VPB	978	7	6.0	1.85	1.97	2.05	0.01
VND	897	6	7.0	2.31	2.35	3.01	0.00
BID	864	10	7.0	1.85	1.95	2.07	0.00
CTG	811	5	7.0	2.13	2.12	2.05	0.00
STB	658	12	8.0	1.95	1.95	2.26	0.00
DIG	535	4	8.5	1.38	1.38	1.35	0.00
GAS	450	12	9.0	1.08	1.25	1.53	0.46
KDH	398	5	10.0	1.32	1.32	1.25	0.20
NLG	398	6	11.0	2.28	2.24	1.57	0.06
MWG	512	8	12.0	1.80	1.82	2.06	0.02
SBT	231	2	12.0	0.88	0.88	—	0.91
TCH	435	4	13.5	2.37	2.09	2.53	0.00
POW	372	5	14.0	1.40	1.40	2.03	0.62
PVD	328	6	17.5	1.74	1.74	2.03	0.09
HSG	225	0	21.0	1.71	1.71	—	0.13
HHV	223	5	22.0	1.30	1.64	—	0.00
DCM	298	6	22.0	1.94	1.93	—	0.30
REE	151	2	44.0	2.65	2.65	—	—

Table 11. Co-trading dependence between assets. "Crude" pools all bins; "M–H" stratifies by time of day. The last two columns give the share of pairs with an odds ratio above one, and the share whose bootstrap interval excludes one.

Market	bin	pairs	activity	median odds ratio crude	M–H	>1 (%)	CI>1 (%)
Binance	1 s	91	0.362	1.08	1.03	80	32
Binance	5 s	171	0.451	1.06	1.04	60	13
Binance	30 s	55	0.436	1.01	1.01	55	11
HOSE	5 s	435	0.368	1.16	1.13	76	27
HOSE	15 s	435	0.657	1.15	1.12	63	15
HOSE	60 s	65	0.826	1.03	0.83	40	2

Table 12. The same co-trading odds ratio computed inside the model, by Mantel–Haenszel over (path, sub-window) strata, read at the observed activity levels. The $a = 0$ row is the independent-clock null.

loading a	model M–H odds ratio at activity 0.362	0.451	0.368
0	1.37	1.87	1.56
0.1	1.56	1.81	1.68
0.25	2.17	2.55	2.37
0.5	2.25	2.86	2.52
1	2.67	3.68	3.02
2	2.85	3.22	3.05
4	3.02	3.09	3.08

Note: The empirical section rejects the $\alpha < 1$ marginal, so this is a magnitude benchmark for the dependence structure, not a fitted model.